

MTH 446: Lecture # 15

Cliff Sun

September 30, 2026

Lecture Span

1. Complex logs

Complex functions

Definition 0.1. $f : \mathbb{C} \rightarrow \mathbb{C}$ is an entire function if $f \in H(\mathbb{C})$.

Here, $\exp(z)$ is an entire function. Moreover, $u, v \in C^\infty$. Also, we can verify

$$\frac{d}{dz} \exp(z) = \frac{d}{dx} \exp(z) = \exp(z) \quad (1)$$

Also, $\exp(z)$ can be negative.

Logarithm

Note, $w = \log z \iff e^w = z$. Note, $\log$ is not defined for $z = 0$. Let $z = r \exp(i\theta)$, then

$$\exp(u + iw) = \exp(u) \exp(iw) = r \exp(i\theta) \quad (2)$$

Therefore,

$$r = \exp(u), \quad \exp(iw) = \exp(i\theta) \quad (3)$$

Therefore,

$$u = \ln(r), \quad w = \theta + 2\pi n, n \in \mathbb{Z} \quad (4)$$

Therefore,

$$\log(z) = \{\ln r + i(\theta + 2\pi n) : n \in \mathbb{Z}\} \quad (5)$$

Or

$$\log(z) = \ln |z| + i \arg(z) \quad (6)$$

Thus, $\log$ is a multi-valued function. By the definition of $\log z$, therefore,

$$\exp(\log(z)) = z \quad (7)$$

Example

Find $\log(\exp(z))$. Note $\exp(z)$. Therefore, $r = \exp(x)$, $\iff \ln(r) = x$. Note,

$$\log(z) = \{x + i(y + 2\pi n) : n \in \mathbb{Z}\} \quad (8)$$

$$= \{z + 2\pi ni : n \in \mathbb{Z}\} \quad (9)$$

Definition 0.2. The Principal value of $\log(z)$ is defined as (capitalized Log) $\text{Log}(z) = \ln|z| + i \arg(z)$. Therefore,

$$\text{Log}(z) = \ln(r) + i\theta, \quad \theta \in (-\theta, \theta] \quad (10)$$

The domain of $\text{Log}(z)$ (principal argument) is considered for all domain except for the line $(-\infty, 0]$. This is because bound of the principle value is from $-\pi < \theta \leq \pi$.

Example

$\log(-1)$? Note, then

$$\log(-1) = \{\ln|-1| + i(\pi + 2\pi n)\} \quad (11)$$

Then, principal value of -1 is $i\pi$.

Definition 0.3. Define the single valued branch of $\log(z)$ as follows:

$$\log(z) = \ln(r) + i\theta, \quad \alpha < \theta < 2\pi + \alpha \quad (12)$$

If $\alpha = -\pi$, then we obtain the principal branch of $\log(z)$.

The purpose of the principal branch is to keep $\log$ a single-valued function.

Example

Let $\alpha = \pi/4$, then

$$\pi/4 < \theta < 9\pi/4 \quad (13)$$

Then find $\log(-1)$. Note that $\text{Log}(-1) = i\pi$ is the principal value of $\log(-1)$. Then, $\text{Log}(-1)$ isn't defined for principal branch when $\alpha = -\pi$, but is define when $\alpha = \pi/4$.

For $1 - i$, $\arg(1 - i)$ is $-\pi/4$, but to find the principal branch,

$$\pi/4 < -\pi/4 + 2\pi n < 9\pi/4 \quad (14)$$

Since $n = 0$ is not defined, try $n = 1$, then we obtain

$$-\pi/4 + 2\pi = 7\pi/4 \quad (15)$$

Therefore, the principal branch of $-\pi/4$ is $7\pi/4$. Therefore,

$$\log(1 - i) = \ln\sqrt{2} + i7\pi/4 \quad (16)$$